Janghyun **Noh**

AI Researcher · Data Scientist

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Education

Myongji University — *Seoul, S.Korea*

M.S. in Artificial Intelligence — *Sep. 2024 - Aug. 2026*

- Thesis: *Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data*
- Advisor: Prof. Minsik Oh, Artificial Intelligence and Data Analytics (AIDA) Lab
- GPA: 4.44 / 4.5
- Award: Research Honor Award from the President of Myongji University (at graduation)
- Teaching Assistant: Deep Learning, Machine Learning, Data Mining, Basic Programming (Fall 2024 – Spring 2026)

Myongji University — *Seoul, S.Korea*

B.S. in Data Technology, School of Convergence Software — *Mar. 2018 - Aug. 2024*

- Capstone Project: CatchYou, a generative AI service that creates composite sketches from witness descriptions
- Leadership: Secretary General, Student Council

RED School – Alternative School — *Geumsan, S.Korea*

Student — *Mar. 2012 - Jan. 2018*

- Scholarship: Academic Excellence Scholarship (two consecutive years)
- Leadership: Student Council President

Experience

Artificial Intelligence and Data Analytics Lab, Myongji University — *Seoul, S.Korea*

Graduate Researcher — *Sep. 2024 - Aug. 2026*

- Developed scCap, a deep learning framework that predicts patient phenotypes from single-cell data without manual cell type annotation, using knowledge-augmented clustering (first-author paper in Briefings in Bioinformatics).
- Developed a multimodal fusion framework for COVID-19 severity prediction that combines clinical records, chest X-rays, and radiology reports and is extensible to new modalities (with KDCA).
- Developed deep learning models that classify tumor subtypes from single-cell data and applied them to drug target discovery (with AIgenDrug).
- Built an LLM-based AI agent environment (Biomni) to automate genomic data analysis for a CRISPR genome-editing analysis platform.
- Set up a scalable genome analysis workflow for apple whole-genome data using the Nextflow nf-core/sarek pipeline (with KNU).
- Analyzed spatial gene-expression data from breast cancer tissue to identify cell neighborhoods and the tumor microenvironment (with SNUH).
- Compared gene-expression patterns between liver cancer tissues with and without cirrhosis using spatial transcriptomics data.

Artificial Intelligence and Data Analytics Lab, Myongji University — *Seoul, S.Korea*

Undergraduate Researcher — *Dec. 2022 - Aug. 2024*

- Developed machine learning models to predict in-hospital mortality of ARDS patients from ICU records, comparing missing-data strategies.
- Identified hypertension-progression risk groups in hyperlipidemia patients from KoGES cohort data by designing lab-based labels and building machine learning and deep learning models (with KDCA).

Bigleader — *Seoul, S.Korea*

AI Intern (Youth AI Internship) — *Jul. 2022 - Oct. 2022*

- Built a supply-chain risk monitoring system that automatically collects foreign customs notices, and validated its keyword extraction and HS code matching (with KOTRA).
- Received the KOTRA Merit Award (President's Award) and the TriBIG Project Scholarship.

Publications & Presentations

Journal Articles

Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data — *Briefings in Bioinformatics (BIB)*

Published — *Jul. 2026*

J. Noh, Y. Shin, M. Kim, and M. Oh

Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids — *Journal of Experimental & Clinical Cancer Research (JECCR)*

Published — *May 2026*

J. Choi, K. Lee, M. Oh, **J. Noh**, Y. Kim, E. Hwang, I. Cho, W. Paik, J. Ryu, J. Ku, and S. Lee

Conference Presentations

Patient-Level Metacell Construction with Pre-trained Embeddings for Phenotype Prediction from scRNA-seq
KIISE Korea Computer Congress (KCC)
Oral Presentation — *Jun. 2026*
Y. Shin, **J. Noh**, and M. Oh

Lightweight Analysis of Patient Phenotype Prediction Based on Single-Cell RNA-seq Data Reduction Techniques
KIISE Korea Software Congress (KSC)
Poster Presentation (Excellent Poster Award) — *Dec. 2025*
Y. Shin, **J. Noh**, and M. Oh

An annotation-free framework for phenotype prediction using hierarchical multiple instance learning with single-cell RNA-sequencing data
International Conference on Genome Informatics ISCB-Asia
Poster Accepted — *Dec. 2025*
J. Noh, Y. Shin, M. Kim, and M. Oh

Patient Phenotype Prediction from Single-cell RNA-seq Data through Annotation-free Multiple Instance Learning
BIOINFO, Korean Society for Bioinformatics (KSBI)
Poster Presentation (Excellent Poster Award) — *Oct. 2025*
J. Noh, Y. Shin, M. Kim, and M. Oh

AMIL: Automated Cell Grouping Framework for Phenotype Prediction with Multiple Instance Learning
KIISE Korea Computer Congress (KCC)
Oral Presentation (Excellent Presentation Award) — *Jul. 2025*
J. Noh, M. Kim, and M. Oh

Predicting In-Hospital Mortality in Patients with Acute Respiratory Distress Syndrome Using MIMIC-III Data: A Comparative Study of Machine Learning Models with Varied Missing Data Handling Strategies
KIISE Korea Software Congress (KSC)
Poster Presentation — *Dec. 2023*
J. Noh, G. Choi, and M. Oh

Honors & Awards

Awards

2026	**Research Honor Award (President's Award),** Myongji University	*Seoul, S.Korea*
2025	**Excellent Poster Award,** KIISE Korea Software Congress (KSC)	*Seoul, S.Korea*
2025	**Excellent Poster Award,** Korean Society for Bioinformatics (KSBI)	*Seoul, S.Korea*
2025	**Excellent Presentation Award,** KIISE Korea Computer Congress (KCC)	*Seoul, S.Korea*
2022	**Merit Award (President's Award),** Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*

Scholarships

2022	**TriBIG Project Scholarship,** Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*
2019	**Globalization Scholarship,** Myongji University	*Seoul, S.Korea*
2018	**Scholarship III,** Myongji University	*Seoul, S.Korea*
2016	**Academic Excellence Scholarship,** RED School	*Geumsan, S.Korea*
2015	**Academic Excellence Scholarship,** RED School	*Geumsan, S.Korea*

Projects

2024 – 2026	CAR-Treg cell therapy for chronic inflammatory disease through an AI-based target discovery platform
2024 – 2026	Genetic and physiological mechanisms of apple blossom dormancy release and flowering under climate change
2024 – 2026	Novel therapeutics for refractory solid tumors through spatial transcriptomics target discovery and organoid validation
2025 – 2026	Client-side web platform for detecting and quantifying large CRISPR-Cas9-induced DNA deletions
2025 – 2026	Pathological differences and immunomodulatory targets in cirrhotic vs. non-cirrhotic HCC (GeoMx)
2025	COVID-19 severity prediction model and environment using multimodal clinical and imaging data
2023 – 2024	Clinical and genetic differences in hypertension-progression risk groups of hyperlipidemia patients (KoGES)
2023	CatchYou: composite sketch generation service based on witness descriptions
2022	Customs notice analysis and monitoring system for global supply-chain crises

Technical Skills

Programming	Python, R, SQL
Frameworks	PyTorch, TensorFlow, Scikit-learn, pandas, NumPy
Tools	Linux, Git, Docker
Research	Machine Learning & Deep Learning, Multimodal Learning, Representation Learning, Predictive Modeling
Languages	Korean (Native), English (Fluent)
Certifications	Engineer Information Processing (2023), SQL Developer (SQLD, 2023)

Extracurricular Activities

Domestic

Google Developer Student Clubs, Myongji University
Seoul, S.Korea

Member — *Oct. 2022 - Dec. 2023*

- Joined the AI team and engaged in machine learning and deep learning study groups
- Collaborated on AI projects and participated in AI competitions

Data Youth Campus, MSIT & KDATA
Seoul, S.Korea

Participant — *Jul. 2022 - Aug. 2022*

- Completed a government-funded data science training program run by the Ministry of Science and ICT and KDATA, hosted at Dongguk University
- Conducted practical machine learning and data analysis projects

DeepsDeeps, Myongji University
Seoul, S.Korea

Member — *Mar. 2022 - Jul. 2022*

- Studied deep learning fundamentals and paper implementations in a peer-led deep learning study group

FAA Football Analytics Academy, Bepro
Seoul, S.Korea

Participant — *Sep. 2021 - Dec. 2021*

- Completed the Data Integration Course on integrating and analyzing football match data

Coding 101, Myongji University
Seoul, S.Korea

Member — *Mar. 2018 - Jan. 2020*

- Studied programming fundamentals and worked on coding projects in a student club

International

Camino de Santiago Pilgrimage
Spain

Pilgrim — *Sep. 2026 - Oct. 2026*

- Walked the French Way on foot from Saint-Jean-Pied-de-Port to Santiago de Compostela

Medical Mission Trip
Manila, Philippines

Volunteer — *Jan. 2024*

- Supported medical volunteer activities with international healthcare professionals through the Global Medical Missions Alliance (GMMA)

International Work Camp
Reykjavik, Iceland

Volunteer — *Dec. 2019 - Jan. 2020*

- Participated in environmental campaigns and intercultural exchange with international volunteers

Short-term Mission Trip
Hanoi, Vietnam

Volunteer — *Aug. 2018*

- Participated in intercultural exchange and community volunteer activities